

Each window will have a small logo in the corner using the file **Small logo.ico**

Login page

Log in page for 5 users.

- The users and passwords can only be changed by the developer.
- Data base with the users must be encrypted

Example of basic login page.

Logo in the middle of the screen and under it, will be user and password insertion. Use **Logo.png**

Pulse ENTER in keyboard to log in with no necessity to click on a button

Button "LOGIN" to log in.

Password not visible.

Selection of the project page

In this screen will appear 2 buttons:

NEW PROJECT – This button will load an empty project

LOAD PROJECT – This button will open Windows file explorer to look for of a project already created

After clicking in any of those buttons will be directed to PROJECT MODULE.

NEW PROJECT

LOAD PROJECT

MASTER SCREEN

It must have a menu at the left side that can be hided.

At the top, the name of the user should appear and at the right side a button to save project.

Button "**save project**" must have an icon and will save all the information of the project in the file created or selected in start screen

Next to it there is a button for "**save as**". It will create a new file with the info of the project.

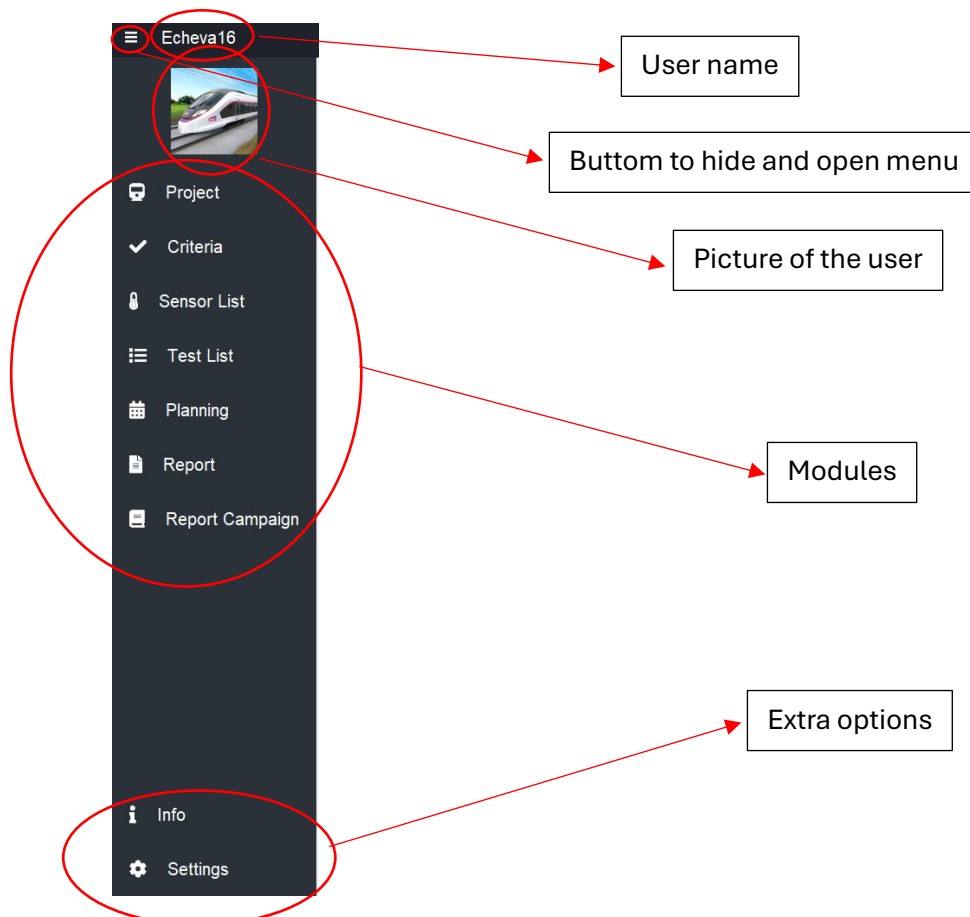


**I dont know what kind of file would be better, could It be a file that only can be opened with the software?like Project.pr?* File must be encrypted.

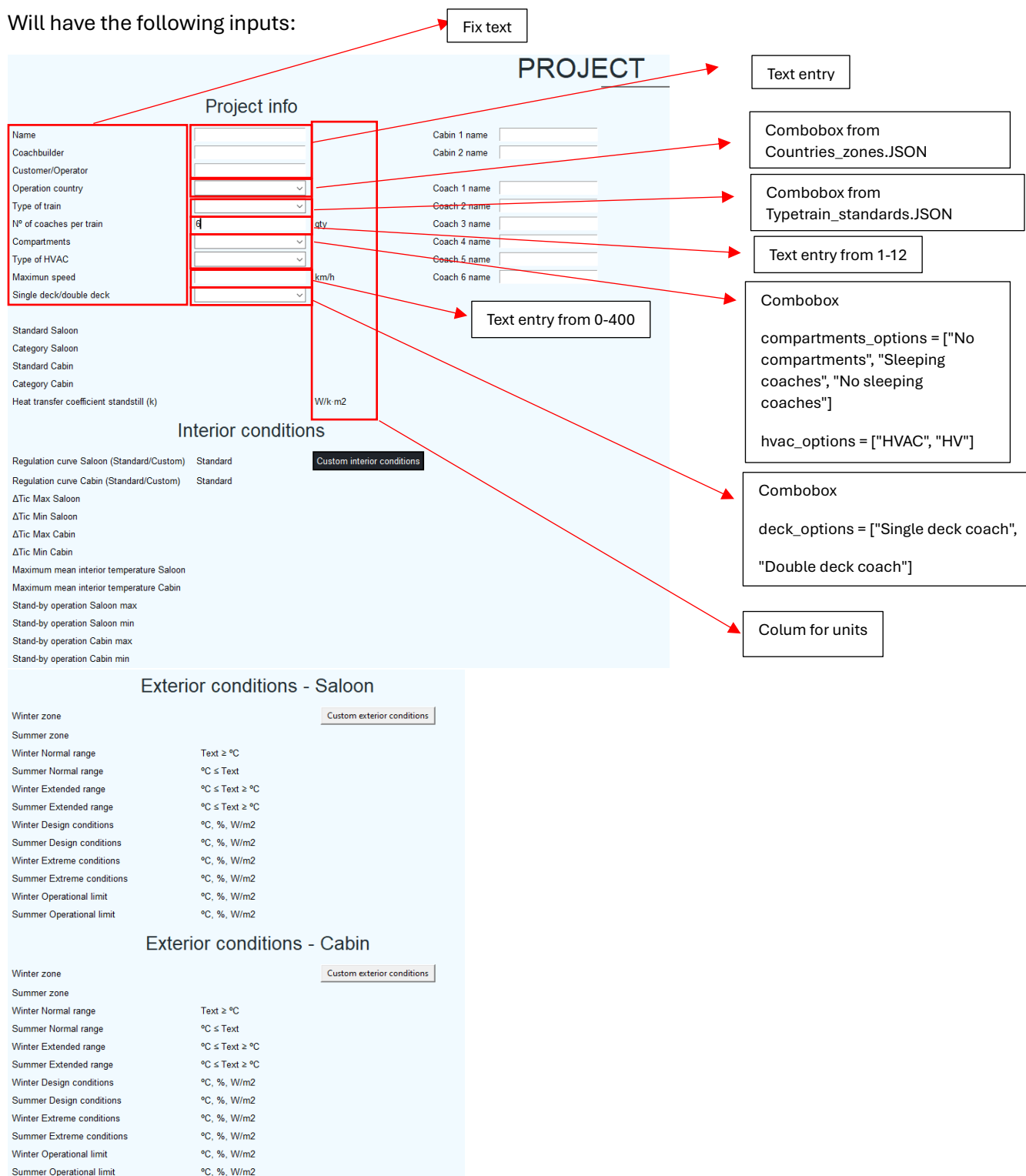
It will contain the following buttons in the left menu.

- PROJECT
 - CRITERIA
 - **COACH DATA*** This does not appear in the picture.
 - SENSOR LIST
 - TEST LIST
 - PLANNING
 - REPORT
 - REPORT CAMPAIGN
-
- INFO
 - SETTINGS

**I dont know what kind of file would be better, could It be a file that only can be opened with the software?like Project.pr?* File must be encrypted.



Will have the following inputs:



PROJECT

Project info

Name
Coachbuilder
Customer/Operator
Operation country
Type of train
N° of coaches per train qty
Compartments
Type of HVAC
Maximum speed km/h
Single deck/double deck

Cabin 1 name
Cabin 2 name
Coach 1 name
Coach 2 name
Coach 3 name
Coach 4 name
Coach 5 name
Coach 6 name

Standard Saloon
Category Saloon
Standard Cabin
Category Cabin
Heat transfer coefficient standstill (k) W/k.m2

Interior conditions

Regulation curve Saloon (Standard/Custom) Standard Custom interior conditions
Regulation curve Cabin (Standard/Custom) Standard
 ΔT_{ic} Max Saloon
 ΔT_{ic} Min Saloon
 ΔT_{ic} Max Cabin
 ΔT_{ic} Min Cabin
Maximum mean interior temperature Saloon
Maximum mean interior temperature Cabin
Stand-by operation Saloon max
Stand-by operation Saloon min
Stand-by operation Cabin max
Stand-by operation Cabin min

Exterior conditions - Saloon

Winter zone Custom exterior conditions
Summer zone
Winter Normal range Text $\geq$ °C
Summer Normal range °C $\leq$ Text
Winter Extended range °C $\leq$ Text $\geq$ °C
Summer Extended range °C $\leq$ Text $\geq$ °C
Winter Design conditions °C, %, W/m2
Summer Design conditions °C, %, W/m2
Winter Extreme conditions °C, %, W/m2
Summer Extreme conditions °C, %, W/m2
Winter Operational limit °C, %, W/m2
Summer Operational limit °C, %, W/m2

Exterior conditions - Cabin

Winter zone Custom exterior conditions
Summer zone
Winter Normal range Text $\geq$ °C
Summer Normal range °C $\leq$ Text
Winter Extended range °C $\leq$ Text $\geq$ °C
Summer Extended range °C $\leq$ Text $\geq$ °C
Winter Design conditions °C, %, W/m2
Summer Design conditions °C, %, W/m2
Winter Extreme conditions °C, %, W/m2
Summer Extreme conditions °C, %, W/m2
Winter Operational limit °C, %, W/m2
Summer Operational limit °C, %, W/m2

The number of coaches are updated when number coaches is inserted in "n° of coaches per

The standard is obtained from Typetrain_standards.JSON with the inputs from "type of train" combobox. In case of "Main line" it is a fix value if not it should appear a bombobox with the options

The category is obtained from Typetrain_standards.JSON with the inputs from "type of train" combobox.and standard.

The standard is obtained from Typetrain_standards.JSON with the inputs from "type of train" combobox.

The category is obtained from Typetrain_standards.JSON with the inputs from "type of train" combobox.and standard.

It will appear heat transfer coefficient SALOON standstill (k) and appear heat transfer coefficient CABIN standstill (k), the info is obtained from k_coefficient.json

PROJECT

Project info

Name		Cabin 1 name	
Coachbuilder		Cabin 2 name	
Customer/Operator			
Operation country		Coach 1 name	
Type of train		Coach 2 name	
N° of coaches per train	qty	Coach 3 name	
Compartments		Coach 4 name	
Type of HVAC		Coach 5 name	
Maximum speed	km/h	Coach 6 name	
Single deck/double deck			

Standard Saloon

Category Saloon

Standard Cabin

Category Cabin

Heat transfer coefficient standstill (k)

W/k.m2

Interior conditions

Regulation curve Saloon (Standard/Custom)	Standard
Regulation curve Cabin (Standard/Custom)	Standard
ΔTic Max Saloon	
ΔTic Min Saloon	
ΔTic Max Cabin	
ΔTic Min Cabin	
Maximum mean interior temperature Saloon	
Maximum mean interior temperature Cabin	
Stand-by operation Saloon max	
Stand-by operation Saloon min	
Stand-by operation Cabin max	
Stand-by operation Cabin min	

Custom interior conditions

This will be standard by default, except if "custom interior conditions" is pushed.

Info obtained from delta_tic_data.json

Info obtained from ti_max.json

Info obtained from standby.json

This button will open a form to overwrite parameters. The parameters overwritten must change the back color to yellow. **See picture of the form**

Exterior conditions - Saloon

Winter zone		Custom exterior conditions
Summer zone		
Winter Normal range	Text ≥ °C	
Summer Normal range	°C ≤ Text	
Winter Extended range	°C ≤ Text ≥ °C	
Summer Extended range	°C ≤ Text ≥ °C	
Winter Design conditions	°C, %, W/m2	
Summer Design conditions	°C, %, W/m2	
Winter Extreme conditions	°C, %, W/m2	
Summer Extreme conditions	°C, %, W/m2	
Winter Operational limit	°C, %, W/m2	
Summer Operational limit	°C, %, W/m2	

Exterior conditions - Cabin

Winter zone		Custom exterior conditions
Summer zone		
Winter Normal range	Text ≥ °C	
Summer Normal range	°C ≤ Text	
Winter Extended range	°C ≤ Text ≥ °C	
Summer Extended range	°C ≤ Text ≥ °C	
Winter Design conditions	°C, %, W/m2	
Summer Design conditions	°C, %, W/m2	
Winter Extreme conditions	°C, %, W/m2	
Summer Extreme conditions	°C, %, W/m2	
Winter Operational limit	°C, %, W/m2	
Summer Operational limit	°C, %, W/m2	

Project info

EN14750:2006

ΔTic Max

ΔTic Min

Custom max

Custom min

EN14813:2006

ΔTic Max

ΔTic Min

Custom max

Custom min

EN14750:2006

Maximum mean interior temperature

°C

°C

EN14813:2006

Maximum mean interior temperature

°C

°C

Stand-by operation

EN14750:2006

Max Saloon

Min Saloon

Custom max

Custom min

EN14813:2006

Max Saloon

Min Saloon

Custom max

Custom min

REGULATION CURVE

Custom/norm

Norm

EN14750:2006

upper limit

lower limit

curve

Custom

upper limit

lower limit

curve

PROJECT

Project info

Name		Cabin 1 name	
Coachbuilder		Cabin 2 name	
Customer/Operator			
Operation country		Coach 1 name	
Type of train		Coach 2 name	
N° of coaches per train	qty	Coach 3 name	
Compartments		Coach 4 name	
Type of HVAC		Coach 5 name	
Maximum speed	km/h	Coach 6 name	
Single deck/double deck			

Standard Saloon	
Category Saloon	
Standard Cabin	
Category Cabin	
Heat transfer coefficient standstill (k)	W/k m2

Interior conditions

Regulation curve Saloon (Standard/Custom)	Standard	Custom interior conditions
Regulation curve Cabin (Standard/Custom)	Standard	
ΔTic Max Saloon		
ΔTic Min Saloon		
ΔTic Max Cabin		
ΔTic Min Cabin		
Maximum mean interior temperature Saloon		
Maximum mean interior temperature Cabin		
Stand-by operation Saloon max		
Stand-by operation Saloon min		
Stand-by operation Cabin max		
Stand-by operation Cabin min		

Exterior conditions - Saloon

Winter zone	
Summer zone	
Winter Normal range	Text ≥ °C
Summer Normal range	°C ≤ Text
Winter Extended range	°C ≤ Text ≥ °C
Summer Extended range	°C ≤ Text ≥ °C
Winter Design conditions	°C, %, W/m2
Summer Design conditions	°C, %, W/m2
Winter Extreme conditions	°C, %, W/m2
Summer Extreme conditions	°C, %, W/m2
Winter Operational limit	°C, %, W/m2
Summer Operational limit	°C, %, W/m2

Custom exterior conditions

Info obtained from conditions_standard.json

Button to open a form to change standard conditions for custom conditions, every custom condition that overwrites any standard must have yellow back. **See picture of the form.**

Exterior conditions - Cabin

Winter zone	
Summer zone	
Winter Normal range	Text ≥ °C
Summer Normal range	°C ≤ Text
Winter Extended range	°C ≤ Text ≥ °C
Summer Extended range	°C ≤ Text ≥ °C
Winter Design conditions	°C, %, W/m2
Summer Design conditions	°C, %, W/m2
Winter Extreme conditions	°C, %, W/m2
Summer Extreme conditions	°C, %, W/m2
Winter Operational limit	°C, %, W/m2
Summer Operational limit	°C, %, W/m2

Custom exterior conditions

Winter zone	EN14813:2006	Custom
Summer zone	II	I
Area	EN14750:2006	Custom
Winter Normal range	Max [°C] Min [°C]	Max [°C] Min [°C]
Summer Normal range	- -	- -
Winter Extended range	- -	- -
Summer Extended range	- -	- -
	EN14813:2006	
Winter Design conditions	Temperature [°C] Relative humidity [%] Solar load [W/m2]	Temperature [°C] Relative humidity [%] Solar load [W/m2]
Summer Design conditions	-20 - 40 40% 800	- - - -
Winter Extreme conditions	- - - -	- - - -
Summer Extreme conditions	- - - -	- - - -
Winter Operational limit	-25 - 40% 800	- - - -
Summer Operational limit	45 - - -	- - - -